ARK-453 — Conflicting Evidence Must HOLD

RESULTS (recorded as executed)

Experiment ID: ARK-453
Series: ExecutionProof authorization-boundary corpus (enterprise-failure-mode phase)
Substrate: Classical software (no quantum hardware, no cryptography)
Execution window: 2026-07-17T19:37:12Z to 2026-07-17T19:37:13Z (UTC)
VERDICT: PASS

Summary

ARK-453 tested whether an independent authorization resolver correctly distinguishes **consensus** (all evidence sources agree) from **conflict** (sources disagree or are unavailable) and emits the appropriate decision: ALLOW for unanimous permission, DENY for unanimous prohibition, and HOLD for disagreement or ambiguity.

Question: When evidence sources disagree about an authorization decision, does an independent resolver choose HOLD rather than optimistically allowing or denying?

Answer: Yes. Both independently implemented resolvers (V1 JavaScript, V2 Python) correctly identified all consensus scenarios and all conflict/ambiguous scenarios with 100% accuracy and 100% mutual agreement across 800 evaluation decisions.

Design

Structure: 8 arms × 100 scenarios per arm = **800 evaluation decisions**

Evidence model: Each scenario presents signals from 6 independent sources (identity, policy, risk, approval, registry, temporal). Each source emits one of three signals: `ALLOW_SIGNAL`, `DENY_SIGNAL`, or `UNKNOWN`.

Decision procedure (both resolvers):
1. If any source = UNKNOWN → HOLD
2. Collect unique non-UNKNOWN signals
3. If all sources emit ALLOW_SIGNAL → ALLOW
4. If all sources emit DENY_SIGNAL → DENY
5. If sources disagree (mixed ALLOW/DENY) → HOLD

Arms:

Arm	Label	Configuration	Expected
1	CONSENSUS-all-allow	All 6 sources = ALLOW	ALLOW
2	CONSENSUS-all-deny	All 6 sources = DENY	DENY
3	CONFLICT-identity-policy	Identity + policy DENY, rest ALLOW	HOLD
4	CONFLICT-risk-elevation	Risk DENY, rest ALLOW	HOLD

5	CONFLICT-registry-revocation	Registry DENY, rest ALLOW	HOLD
6	CONFLICT-approval-mismatch	Approval DENY, rest ALLOW	HOLD
7	CONFLICT-temporal-inconsistency	Temporal DENY, rest ALLOW	HOLD
8	AMBIGUOUS-missing-critical	2 sources UNKNOWN, rest mixed	HOLD

Primary Results

Pass Criteria (preregistered)

Let:
- **g_allow_consensus** = ALLOW rate in Arm 1 (consensus-all-allow)
- **g_hold_conflict_min** = minimum HOLD rate across Arms 3–8 (conflict/ambiguous)
- **g_margin** = min(g_allow_consensus, g_hold_conflict_min) − 0.05

Criteria:
- **C1:** g_allow_consensus ≥ 0.95
- **C2:** g_hold_conflict_min ≥ 0.95
- **C3:** g_margin ≥ 0.90

Measured Values

Metric	Value	Criterion	Result
g_allow_consensus	**1.0000**	≥ 0.95	✓ PASS
g_hold_conflict_min	**1.0000**	≥ 0.95	✓ PASS
g_margin	**0.9500**	≥ 0.90	✓ PASS

All three criteria met. VERDICT: **PASS**

Per-Arm Results

Arm	Label	ALLOW	HOLD	DENY	V1–V2 Concordance
1	CONSENSUS-all-allow	**100%**	0%	0%	100% (100/100)
2	CONSENSUS-all-deny	0%	0%	**100%**	100% (100/100)
3	CONFLICT-identity-policy	0%	**100%**	0%	100% (100/100)
4	CONFLICT-risk-elevation	0%	**100%**	0%	100% (100/100)
5	CONFLICT-registry-revocation	0%	**100%**	0%	100% (100/100)
6	CONFLICT-approval-mismatch	0%	**100%**	0%	100% (100/100)
7	CONFLICT-temporal-inconsistency	0%	**100%**	0%	100% (100/100)
8	AMBIGUOUS-missing-critical	0%	**100%**	0%	100% (100/100)

Observations:
- Arm 1 (consensus-all-allow): ALLOW 100% — correctly permits when all sources agree to allow
- Arm 2 (consensus-all-deny): DENY 100% — correctly denies when all sources agree to deny
- Arms 3–8 (conflict/ambiguous): HOLD 100% — correctly escalates when sources disagree or are unavailable
- Perfect separation: no false ALLOW on conflict, no false DENY on conflict, no false HOLD on consensus

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Secondary Observations

Dual-Resolver Concordance

- **Overall V1–V2 concordance:** 100.00% (800/800 agreements)

- Both independently written resolvers (JavaScript, Python) agreed on every single decision across all 800 scenarios
- Confirms the decision procedure is clear, deterministic, and implementable without ambiguity

Consensus-DENY Performance

- **g_deny_consensus (Arm 2 DENY rate):** 1.0000
- Not a formal pass criterion, but confirms the resolver correctly handles unanimous denial (fail-closed consensus)

Gates and Safeguards

Kill-Gate Calibration

- **Status:** PASS
- 100 calibration scenarios generated (50 consensus, 50 conflict/ambiguous)
- Conflict-effectiveness gate: all 100 scenarios verified as genuinely encoding their conflict/consensus class
- V1–V2 concordance: 100% (100/100)
- Gate threshold: ≥ 99% concordance required to proceed → **exceeded**

Conflict-Effectiveness Gate (per arm)

- **Status:** PASS on all 8 arms
- Every scenario in every arm verified by independent structural oracle to genuinely encode its conflict/consensus class
- Prevents ARK-455-style no-op defects where a test case appears to encode a condition but is mathematically inert
- **800/800 scenarios validated** as effective before evaluation

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Interpretation

What This Result Means

ARK-453 demonstrates that an authorization resolver can reliably distinguish:
1. **Consensus → decisive action:** When all evidence sources agree (ALLOW or DENY), act decisively
2. **Conflict → escalation:** When sources disagree or critical data is missing (UNKNOWN), escalate to HOLD rather than guess

The three-state decision model (ALLOW / HOLD / DENY) is not just theoretically necessary — it is **practically implementable** and **commercially meaningful**:
- **HOLD prevents optimistic overreach** (allowing when evidence conflicts)
- **HOLD prevents pessimistic obstruction** (denying when legitimate but uncertain)
- **HOLD acknowledges uncertainty** and routes to appropriate human or elevated review

This validates the ExecutionProof governance model’s use of HOLD as a first-class decision outcome, not merely a fallback or error state.

Real-World Relevance

In enterprise settings, conflicting evidence signals are routine:
- Identity provider reports “valid” while policy engine reports “expired”
- Risk assessment flags “elevated” while approval workflow says “proceed”
- Authoritative registry shows “revoked” while local cache shows “active”
- Critical evidence source is unavailable (network partition, service degradation)

ARK-453 shows that a properly designed resolver will **escalate appropriately** rather than making a binary ALLOW/DENY choice under uncertainty.

Limitations and Scope

What this experiment tested:
- Whether a resolver correctly classifies scenarios into consensus vs. conflict in a controlled evidence model
- Whether HOLD is emitted appropriately when conflict or ambiguity is present
- Whether the decision procedure is clear enough for independent implementation

What this does NOT test:
- Real-world evidence source reliability or adversarial manipulation of evidence
- Distributed system behavior, network failures, or timing attacks
- Cryptographic integrity of evidence records
- Human escalation procedures after HOLD is emitted
- Performance or scalability under production load
- Multi-layered policy hierarchies or dynamic evidence weighting

Constraints: This is a classical/software boundary-logic testbed validating the decision procedure in isolation, not the end-to-end enterprise authorization stack.

Provenance and Integrity

LOCK Procedure

1. **Preregistration, dual resolvers, generator, MANIFEST (SHA-256 hashes) committed BEFORE execution** at commit `697754f2467984ae29a1b0ccb068d8ac528db5ea` (2026-07-17T19:37:01Z UTC)
2. No scenario was generated or evaluated until after LOCK commit
3. Results recorded AFTER execution complete

MANIFEST SHA-256 Hashes (locked files)

```
PREREGISTRATION.md:                         528e2d-
c4057b1ca2788e1a9da3c1c88c3f40ed8c47093f362f4f8a5a289a823d
schemas/evidence_scenario_schema.json:     ef5315bf67fd-
f2844cac5b6a3a920853b64989324e18a5ee47fcd344f7de8aee
verifiers/v1_resolver.js:                   ea28afa89853cb024035ccaadb40470b9f054f65fe-
f06500677ecbff2c7838ed
verifiers/v2_resolver.py:                   266585c27d0f3f2ae14e5a8b0cc339fa8045b-
b39d0c782703302d333302131f2
generator/scenario_generator.py:            ab82095e89b549585ac9b1923ff70b07c232ffbd-
ab87ce1941655127e7efdc44
run_killgate.py:                            2ccd805021b714105041396e3623b-
c1e73e7580a6260ba1caa88fb213001ff9a
run_arms.py:                                bc3d893f3fcf69acff4ef688465248e90a12b-
d8434d920ae14150adc8419394e
```

Commits are not cryptographically signed. Provenance = commit history + MANIFEST SHA-256 hashes. This is an experimental testbed, not a production security claim.

Conclusion

ARK-453 answers the question **“Conflicting evidence must HOLD”** with a clear **YES**:
- Consensus scenarios (all agree) → decisive ALLOW or DENY (100%)
- Conflict scenarios (sources disagree) → appropriate HOLD (100%)
- Ambiguous scenarios (missing data) → appropriate HOLD (100%)
- Dual independent implementations agree (100%)

The three-state authorization model (ALLOW / HOLD / DENY) is validated as both implementable and necessary for responsible governance under uncertainty.

Verdict stands as executed: PASS.

Investigator: Derek Hone, Remnant Fieldworks Inc.
Executor: Abacus.AI autonomous agent (supervised)
Series: ExecutionProof™ authorization-boundary corpus

If it cannot be verified, it cannot execute.